

AP Calc Lesson 2.5 Homework

Name _____

1. $\frac{d}{dx}(4 \cos x + 9x) =$

2. $\lim_{x \rightarrow \frac{5\pi}{6}} \frac{\sin x - \frac{1}{2}}{x - \frac{5\pi}{6}}$

3. Let f be a differentiable function given by $f(x) = \begin{cases} b + ax & x > \frac{\pi}{4} \\ \sqrt{2} \cos x & x \leq \frac{\pi}{4} \end{cases}$, where a and b are constants.
Find the values of a and b .

4. Let $f(x) = x - 2 \sin x$.

a. Find $f'(x)$.

b. Write an equation for the line tangent to f at $x = \pi$.

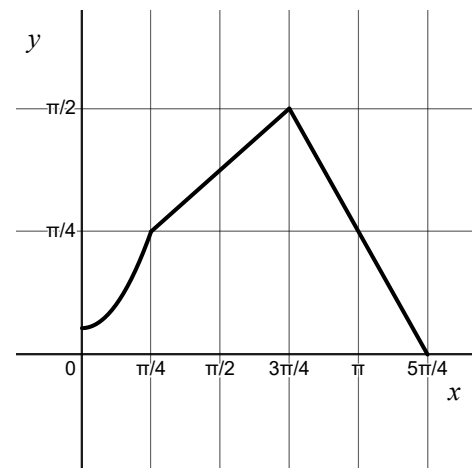
c. Find all values of x on the closed interval $[0, 2\pi]$ such that the line tangent to the graph of f is horizontal.

5. The graph of a function $y = g(x)$ is shown. Let $f(x) = \sin x + g(x)$.

a. Find $f'(\pi)$.

b. Is the graph of f' positive or negative on the interval $(0, \frac{\pi}{4})$? How do you know?

c. Find the value of x on the interval $[\frac{\pi}{4}, \frac{3\pi}{4}]$ such that the line tangent to the graph of f is parallel to the x -axis.



6.

$f(x) = 4x^2 - \sqrt[3]{x}$	$g(x) = \cos x - 4$	$h(x) = \sqrt{3} \sin x$
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The table above gives functions f , g , and h .

a. Write an equation for the line tangent to the graph of g at $x = \frac{\pi}{6}$.

b. Find the value of x on the interval $[0, \pi]$ for which the line tangent to g is parallel to the line tangent to h .

c. Let $j(x) = f(x) - 2g(x)$. Find $j'(x)$.

7.

x	$f(x)$	$f'(x)$
0	3	-4
π	3π	1

Selected values of a continuous function f and its derivative f' are given in the table.

Function g is given by $g(x) = 3 \sin x + 4x - \pi$.

a. Let $h(x) = \frac{f(x)}{2} - g(x)$. Find $h'(0)$.

b. If $j(x) = \begin{cases} f(x) & x \leq \pi \\ g(x) & x > \pi \end{cases}$, determine whether j is differentiable at $x = \pi$. Show the work that leads to your solution.



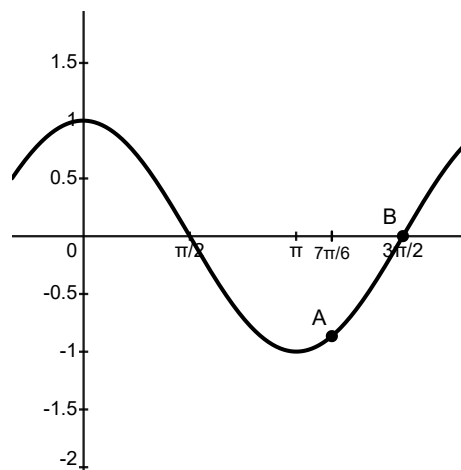
8. Let f be the function defined by $f(x) = \frac{1}{\pi} \sin x - 2$ on the interval $[-4, 4]$. The line tangent to the graph of f at $x = d$ passes through the point $(0, -3)$.

a. Find $f'(x)$.

b. Write, but do not solve, an equation that could be used to determine the value of d .

c. Use technology to find the value of d .

9. The graph of $y = f(x)$ is shown. Which is greater: the slope of f at Point A or the slope of f at Point B? How many times greater?



10. Evaluate $\lim_{h \rightarrow 0} \frac{2 \cos \left(\frac{\pi}{3} + h \right) - 1}{h}$

A) $-\sqrt{3}$

B) 0

C) $\frac{1}{2}$

D) The limit does not exist.

11. The line $y = 4x - 6\pi$ is tangent to the graph of f at $x = a$. Which of the following correctly identifies function f and the value of a ?

A) $f(x) = -4 \sin x$ and $a = \frac{\pi}{2}$

B) $f(x) = -4 \cos x$ and $a = \frac{\pi}{2}$

C) $f(x) = 4 \sin x$ and $a = \frac{3\pi}{2}$

D) $f(x) = 4 \cos x$ and $a = \frac{3\pi}{2}$